

Business Sales Performance Analytics Report

Future Interns – Data Science & Analytics (Task 1)

Prepared By

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1. Executive Summary

This project analyzes business sales performance using the Sample Superstore dataset. The objective was to identify revenue trends, top-performing products, profitable categories, and high-performing regions to support data-driven business decision-making.

The analysis was conducted using Python for data cleaning and preprocessing and Power BI for interactive dashboard development and visualization.

2. Project Objective

The primary objectives of this analysis were:

- Analyze sales performance over time.
 - Identify top revenue-generating products.
 - Evaluate category-wise sales and profitability.
 - Assess regional performance.
 - Generate actionable business recommendations.
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3. Dataset Overview

The analysis was performed using the Sample Superstore Dataset containing:

- Customer information
- Product information
- Order details
- Sales revenue
- Profit

- Discount data
- Regional information

Total Records Analyzed: 9,994

4. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Jupyter Notebook
- Power BI

5. Key Findings

Revenue Trend Analysis

Sales revenue demonstrated consistent growth throughout the observed period.

Year Sales Revenue

2014 \$484,247

2015 \$470,533

2016 \$609,206

2017 \$733,215

Key Observation:

- Revenue increased significantly between 2015 and 2017.
- Overall growth exceeded 50% across the four-year period.

Category Performance Analysis

Sales by Category

1. Technology – \$836,154
2. Furniture – \$741,999
3. Office Supplies – \$719,047

Profit by Category

1. Technology – \$145,455
2. Office Supplies – \$122,491
3. Furniture – \$18,451

Key Observation:

Technology was the strongest-performing category in both revenue and profit generation.

Furniture generated substantial sales but produced comparatively low profit margins.

Regional Performance Analysis

Sales by Region

1. West – \$725,458
2. East – \$678,781
3. Central – \$501,240
4. South – \$391,722

Profit by Region

1. West – \$108,418
2. East – \$91,523
3. South – \$46,749
4. Central – \$39,706

Key Observation:

The West region consistently outperformed all other regions in both revenue generation and profitability.

Top Revenue-Generating Products

Top-performing products included:

1. Canon imageCLASS 2200 Advanced Copier
2. Fellowes PB500 Electric Punch Plastic Comb Binding Machine
3. Cisco TelePresence System EX90 Videoconferencing Unit
4. HON 5400 Series Task Chairs for Big and Tall
5. GBC DocuBind TL300 Electric Binding System

Key Observation:

A small number of premium products contributed significantly to overall revenue.

6. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Expand Technology Product Portfolio

Technology products generated the highest sales and profits, indicating strong customer demand and profitability.

2. Review Furniture Pricing Strategy

Furniture sales are high but profitability remains low. Discounting and pricing policies should be reviewed.

3. Strengthen High-Performing Regions

The West region demonstrated exceptional performance. Similar sales strategies should be replicated in other regions.

4. Prioritize High-Revenue Products

Ensure inventory availability and targeted marketing for top-performing products.

5. Monitor Profitability Alongside Revenue

Business decisions should consider both sales growth and profit margins to maximize long-term value.

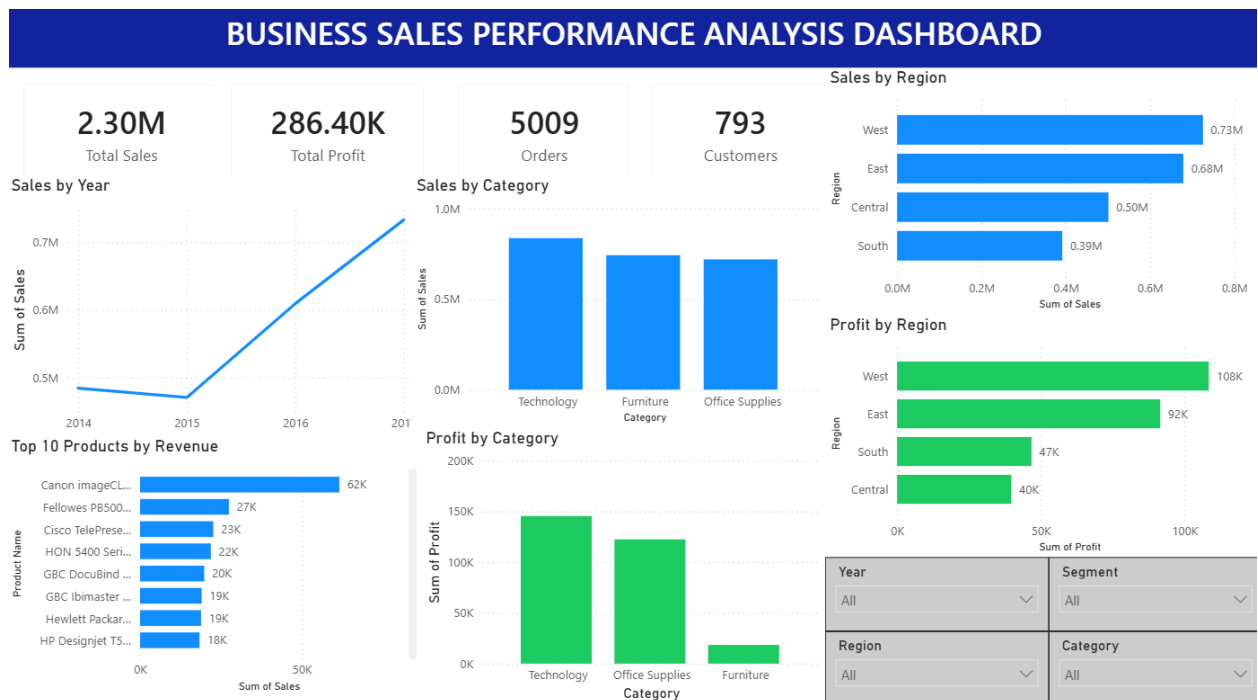
7. Dashboard Deliverable

An interactive Power BI dashboard was developed to visualize:

- Sales trends over time
- Category performance
- Regional performance
- Top-selling products
- Key business KPIs

Dashboard KPIs:

- Total Sales: \$2.30M
- Total Profit: \$286.40K
- Orders: 5,009
- Customers: 793



8. Conclusion

The analysis revealed strong business growth, with Technology emerging as the most profitable category and the West region delivering the strongest performance. By focusing on high-performing products, improving low-margin categories, and leveraging successful regional strategies, the business can further improve profitability and sustain long-term growth.